



Software Engineering Department
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Final Project — Phase A (61998)

Multi-Modal LLM Style Analysis

**Detecting Misattributed Creations in the Shakespeare Corpus through LLM-Based
Style Normalization**

CODE: 26-2-R-6

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Link to GitHub:

<https://github.com/eliranmeleh/Multi-Modal-LLM-Style-Analysis>

Table of Contents

1. Abstract	3
2. Introduction	3
2.1 Background and motivation	3
2.2 Problem statement	4
2.3 Proposed approach and project goals.....	4
3. Literature Review	5
3.1 Computational authorship attribution.....	5
3.2 The Impostors method	5
3.3 Deep learning for authorship	6
3.4 Large Language Models and style transfer	6
3.5 Anomaly detection and threshold selection	7
4. Project Description	8
4.1 Project requirements	8
4.1.1 Functional requirements.....	8
4.1.2 Non-functional requirements.....	8
4.2 Requirements gathering.....	9
4.3 Research process.....	9
4.4 Expected challenges	10
4.5 Tools and technologies.....	10
4.6 Solution description	11
4.6.1 Pipeline overview	11
4.6.2 Pseudocode	13
4.6.3 Function-word edit distance	14
4.7 Worked example: Hamlet end-to-end	15
4.8 Success metrics	15
4.9 Testing process.....	16
4.10 Risk analysis and mitigations.....	17
5. AI Tools and Prompts Used	18
5.1 Claude (Anthropic): methodology refinement.....	18
5.2 The pipeline LLM: style extraction and rewriting.....	18
6. References	20

1. Abstract

The Shakespeare authorship question (whether all creations traditionally attributed to William Shakespeare were indeed written by him) has been the subject of literary and computational debate for over a century. Computational stylometry has long approached the question with classifier-based methods, which typically depend on curated corpora of comparison authors, require one trained classifier per author pair, and produce opaque numerical scores that literary scholars cannot inspect directly.

This project, Multi-Modal LLM Style Analysis, proposes a complementary approach driven by LLM-based style normalization. A large language model first extracts a generic style profile from the 49-creation corpus, then rewrites each creation under that profile while preserving content. A Function-Word Edit Distance between original and rewritten text (one of several distance measures compared in Phase B) measures how much stylistic change was required: authentic Shakespeare chunks need little change; chunks by other authors change substantially. Otsu’s 1-D threshold separates the two groups. Robustness comes from averaging per-creation scores over several independent runs; in the main proposal each run after the first injects one different control “noise” creation into the corpus (non-cumulative), while Phase B also evaluates a variant with no added noise.

The approach is fully unsupervised and zero-shot. It requires no external corpus of comparison authors and no classifier training, produces an interpretable per-creation score (“about 20% of function words changed”), and generalizes to any author’s corpus by direct substitution. The project is expected to deliver a reproducible, open-source implementation, along with initial evidence that the same pipeline transfers to other authorship questions.

2. Introduction

2.1 Background and motivation

Since shortly after Shakespeare’s death, scholars have debated whether the creations attributed to him were indeed his alone. The 2011 film “Anonymous” brought this debate to a wide audience by dramatizing the Oxfordian theory, which posits Edward de Vere as the true author. Scholarly hypotheses range from outright alternative authorship (Bacon, Marlowe) to collaborative attribution (Shakespeare with Fletcher, Middleton, or Marlowe). Modern textual scholarship (most notably the 2016 New Oxford Shakespeare) explicitly credits Christopher Marlowe as co-author of the Henry VI plays, marking a significant shift in mainstream attribution practice. While the Stratfordian consensus remains dominant, the question of co-authorship has become increasingly accepted, supported by both manuscript evidence and quantitative stylometric analysis.

Computational stylometry offers an objective lens on the question. It examines quantifiable features of writing, such as function-word frequencies, sentence-length distributions, and n-gram patterns, to detect authorship signatures. The field was firmly established by Mosteller and Wallace (1964) in their landmark analysis of the disputed Federalist Papers and has since matured into a well-established branch of computational linguistics (Stamatatos, 2009).

2.2 Problem statement

Computational stylometry has approached the authorship question primarily through classifier-based methods. The classical Impostors method (Koppel and Winter, 2014) verifies authorship by comparing a candidate document against a pool of impostor texts (creations by authors other than the target) to test whether the candidate matches the target’s style more closely than the impostors do. Recent extensions, including the Deep Impostor methodology proposed by Volkovich and Avros (2025), replace the original method’s feature-randomization step with deep neural networks (a CNN or a fine-tuned BERT) that perform binary classification of text batches against impostor pairs. On the 49-creation Shakespeare corpus, Deep Impostor identifies approximately 15 creations as suspicious, including well-known co-authorship cases such as Henry VIII (with Fletcher) and Henry VI (with Marlowe), and several plays in the Shakespeare Apocrypha.

These classifier-based approaches are effective, but they share three open challenges that motivate the complementary approach pursued here. They require an external corpus of comparison authors, so applying a method to a different target author means collecting a new pool. They also train one or many neural classifiers (over a hundred in the Deep Impostor study) at substantial computational cost, and so are not zero-shot. And the resulting score is opaque: there is no way to tell a literary scholar, in stylometric terms, why a given creation was flagged.

2.3 Proposed approach and project goals

This project proposes a complementary methodology that uses a modern Large Language Model (LLM) as a style-normalization engine. The LLM extracts a generic style profile from the corpus, then rewrites individual chunks of every creation to match that profile while preserving content. The change measured between original and rewritten chunks, restricted to function-word usage, is the stylistic distance. In practical terms it is the share of a chunk’s function words the model had to alter to bring it in line with the corpus style. The Function-Word Edit Distance is the metric the method is developed with, but the project deliberately takes a multi-modal view of style: it treats it as one of several complementary stylistic representations that can be compared or combined: the edit-distance metric alongside frequency-vector and signal-based distances (Section 4.6.3). This project continues the line of research of Volkovich and Avros (2025): it studies the same corpus and benchmarks against their published results, exploring an unsupervised, LLM-based alternative to their trained-classifier approach. Although the method is demonstrated on the Shakespeare corpus, nothing in it is Shakespeare-specific. Shakespeare is simply a running example, and the same pipeline applies unchanged to any author’s corpus.

Modern LLMs are well suited to this task. They have learned the statistical regularities of style across vast volumes of text and can perform zero-shot style transfer with prompt-only configuration. They require only a corpus and a prompt, with no per-task training, no fine-tuning, no parameter tuning. Their main limitation, possible memorization of well-known texts such as Shakespeare’s, is addressed through the control experiment described in Section 4.3.

The goals of this project are:

- To design and implement a fully unsupervised, zero-shot authorship-anomaly detection pipeline driven by LLM-based style normalization.

- To produce results that are directly interpretable for literary scholars, readable in plain stylistic terms rather than as an opaque score.
- To benchmark the pipeline against existing computational stylometry findings on the Shakespeare corpus, providing a methodological cross-check across complementary approaches to the same problem.
- To show that the methodology generalizes: the same pipeline applies to any author’s corpus by direct substitution.
- To deliver an open-source, reproducible implementation that other researchers can reuse and extend.

3. Literature Review

3.1 Computational authorship attribution

Authorship attribution identifies the author of an unknown text given samples of candidate authors’ writing. The field was firmly established by Mosteller and Wallace (1964), whose Bayesian analysis of the disputed Federalist Papers demonstrated that statistical treatment of function-word frequencies could resolve longstanding authorship disputes. Stamatatos (2009) provides a comprehensive modern survey, distinguishing between profile-based methods (comparing aggregated stylistic features) and instance-based methods (treating each text as an independent training sample).

Function words have proved particularly powerful as a stylistic feature: they are content-independent, used unconsciously by writers, and characteristic of individual style. Successful methods in this tradition compute z-score-based distances over function-word frequencies. The method follows the same tradition; the core metric in the pipeline is the Function-Word Edit Distance between an original chunk and its LLM-rewritten version.

3.2 The Impostors method

Koppel and Winter (2014) introduced the Impostors method, which has become a central technique in the authorship verification literature. The setup is the following. A target author is given (with samples of their known writing), along with a candidate document of disputed authorship. To test whether the target wrote the candidate, the method assembles a pool of impostor texts (documents written by other authors, ideally of similar genre and historical period) that serve as a “background population” of styles that could plausibly be confused with the target’s. The candidate is then compared to both the target’s known writings and the impostor pool across many iterations, each iteration using a randomly selected subset of stylistic features and a randomly selected subset of impostors. If the candidate is consistently more similar to the target than to the impostors across these iterations, attribution to the target is supported. The impostors thus help distinguish a true stylistic match from a coincidental similarity in a randomly chosen feature subspace.

The Impostors method has been particularly successful in PAN authorship-verification competitions and has been extended in numerous follow-up works, including the General Impostors framework, modifications for short texts, and applications to detecting pseudepigraphic creations. Volkovich and Avros (2025) build directly on this lineage, replacing the feature-subset random sampling of the original method with a deep neural network trained to discriminate between impostor pairs. This gives a more expressive comparison than feature-subset cosine similarity, but it still inherits the dependency on an external impostor pool.

3.3 Deep learning for authorship

Recent work has integrated deep neural networks into authorship attribution. Convolutional neural networks (CNNs) over word embeddings have been used successfully for short-text authorship classification, where their ability to capture local n-gram-like patterns is well suited to stylistic discrimination. Transformer-based models, particularly BERT (Devlin et al., 2019), use attention mechanisms to capture longer-range dependencies and contextual word usage; fine-tuning a pre-trained BERT on an authorship task can leverage this rich linguistic representation.

Volkovich and Avros (2025) provide a recent application of these architectures to the Shakespeare corpus, using both within a five-stage Deep Impostor pipeline. First, each tested creation and each impostor creation is divided into batches of $L = 50$ words. Second, a neural classifier is trained to discriminate between batches drawn from two specific impostors; this is either a CNN built from scratch (three 1D-convolutional layers with kernels of 3, 6, 12 words, followed by a bidirectional LSTM and dense classifier layers) or a fine-tuned BERT. Third, each tested document is transformed into a real-valued signal by averaging classifier outputs within chunks of $k = 8$ batches. Fourth, DTW distance is computed between all pairs of signals, a Kuratowski-type embedding maps the resulting matrix back to Euclidean space, and Isolation Forest assigns anomaly scores. Fifth, K-medoids clustering produces the final binary partition into authentic and suspicious creations.

The CNN variant of this pipeline identifies 15 of the 49 canonical Shakespeare creations as suspicious. Several of these align with established scholarly consensus on co-authorship (Henry VI / Marlowe; Timon of Athens / Middleton); others are more speculative. The BERT variant produces a smaller suspicious set (8 creations), which the authors attribute to insufficient post-training and identify as future work.

The methodology takes a different route to leveraging deep models. Rather than training a classifier from scratch or fine-tuning a transformer, it uses an off-the-shelf large language model in a zero-shot setting. The LLM is not asked to perform classification; it is asked to perform style transfer, with the resulting stylistic distance serving as the basis for classification.

3.4 Large Language Models and style transfer

Modern Large Language Models, including GPT-4, Claude, and Gemini, show strong capabilities in style-aware text generation. Trained on internet-scale text corpora, they have learned the statistical regularities of style across a wide variety of authors, genres, and historical periods. Style transfer, rewriting text to match a target style while preserving meaning, has emerged as a productive zero-shot application of LLMs. With appropriate prompting, current models produce transformations competitive with dedicated style-transfer models.

Recent work has begun to apply Large Language Models directly to authorship analysis. Huang et al. (2024) demonstrate that an LLM can perform zero-shot authorship verification and attribution end-to-end, providing linguistic-feature explanations of its decisions. Two further works are methodologically closer to the approach proposed here. Miralles-González et al. (2025) introduce an unsupervised framework that uses log-probabilities of an LLM to measure style transferability between two texts. Their method outperforms unsupervised prompting baselines on the PAN authorship-verification benchmarks, but it does not rewrite the text; it measures probability directly. Icard et al. (2026) are closer still: three LLMs (GPT-4o, Mistral-Large, Gemini-1.5-Flash) rewrite literary texts in the style of Proust, Céline, and Yourcenar, and the embedding dispersion between original and rewritten texts is measured as a stylistic distance. This project differs from these in two respects: the metric is Function-Word Edit Distance, which is interpretable for non-technical readers, and the design explicitly targets a contaminated corpus, with optional iterative purification deferred to Phase B.

LLM-based methods carry a known risk: memorization of widely-circulated training data. Shakespeare’s creations are likely present in the training corpora of all major LLMs, so the model may reproduce its memorized version rather than genuinely normalizing a passage via the given profile. Recent work has shown encouragingly that authorship-verification methods remain resilient against LLM-imitated text, but that finding is not conclusive for this setting. The concern is addressed through the control experiment described in Section 4.3.

3.5 Anomaly detection and threshold selection

Anomaly detection identifies data points that deviate from the bulk of a distribution. Liu, Ting, and Zhou (2008) introduced Isolation Forest, an unsupervised tree-ensemble method that operates on the principle that outliers are easier to isolate via random feature splits than typical points. It is the classifier used in the original Volkovich and Avros (2025) pipeline. However, Isolation Forest assumes a specific data shape: a large “normal” cluster with rare anomalous outliers. When the data contains two clusters of comparable size (as in this case, where approximately 30% of the creations are suspicious), the framework is somewhat misaligned, because the suspicious creations do not behave like rare outliers; they form a second population of comparable size.

For one-dimensional outputs in such a two-cluster setting, simpler thresholding methods are appropriate. Otsu (1979) developed an automatic threshold-selection method that maximizes between-class variance in a 1-D distribution; originally designed for image segmentation, it is fully parameter-free. The method uses Otsu’s threshold on the per-creation mean delta, with a visual sanity check against the gap in the sorted scatter plot. Agreement between Otsu’s threshold and this visible gap provides strong evidence that the two-cluster structure is real.

4. Project Description

4.1 Project requirements

4.1.1 Functional requirements

- Process the canonical 49-creation Shakespeare corpus, sourced from Project Gutenberg (Volkovich and Avros, 2025).
- Extract a generic style profile from the full corpus using an LLM (the majority of the corpus is assumed to be in the author’s style), with no Shakespeare-specific cues in the prompt. Feed whole creations; if the corpus does not fit the LLM’s context window in a single call, split into multiple calls, each holding complete creations.
- Partition every creation into consecutive same-size chunks (default $P = 400$) with full coverage and no sampling.
- Run M independent runs of the pipeline; in the main proposal each run after the first adds one different foreign creation to the corpus as noise; run 1 uses the plain corpus and noise never accumulates across runs (Phase B also tests with no added noise). In run i : extract profile $_i$ from that run’s corpus, rewrite every chunk under profile $_i$, and compute per-creation scores. After the M runs, average each creation’s M scores into a single score.
- Compute Function-Word Edit Distance between every chunk and its rewrite, using a fixed list of about 120 function words covering both modern and Early Modern forms, with symmetric normalization $\delta = \frac{\text{lev}(fW_{\text{orig}}, fW_{\text{rew}})}{\max(|fW_{\text{orig}}|, |fW_{\text{rew}}|)}$.
- Classify creations as authentic or suspicious via Otsu’s threshold on the M -averaged per-creation scores.
- Optionally, and only if the single-round threshold proves unstable, purify the profile by re-extracting it from the authentic-only set and repeating Steps 1 through 6 (this iterative purification is deferred to Phase B), bounded by $T_{\text{max}} = 5$ iterations.
- Run a control experiment comparing rewrites with the explicit profile against rewrites with a generic “in Shakespeare’s style” prompt.
- Produce a sorted scatter plot of all 49 scores with the Otsu threshold marked, plus a labeled list of suspicious creations and their continuous scores. Report per-creation standard deviation across the M runs as a confidence diagnostic.

4.1.2 Non-functional requirements

- **Reproducibility.** Classification labels (authentic vs. suspicious) are stable across all M independent runs for every creation in the corpus, except for the small set of creations independently identified as borderline or mixed-authorship by scholarly consensus (Henry VIII, Pericles, Henry VI Parts I–III, Timon of Athens). The criterion is quantitative and binary at the creation level: any non-borderline creation whose label varies across the M runs counts as a stability failure. Counts of passing and failing creations are reported in the final-phase writeup. This grounds the reproducibility threshold in independent literary scholarship rather than an arbitrary numerical bound.

- **Generalizability.** Pipeline parameters are expressed in terms of creation length, allowing the corpus to be swapped without code changes. No code path may hard-code “Shakespeare.”
- **Interpretability.** Every per-creation score is traceable back to its constituent chunk deltas, and the output must be understandable to non-ML literary readers.
- **Documentation.** All code, prompts, configuration, and result files are committed to a public Git repository (<https://github.com/eliranmeleh/Multi-Modal-LLM-Style-Analysis>).
- **Modularity.** The LLM provider is configurable via a single setting, allowing substitution of Gemini, GPT-4o, or Claude without algorithm changes.
- **Auditability.** Every LLM call is logged with prompt, response, model version, and timestamp, so a run can be reproduced exactly.

4.2 Requirements gathering

The project’s requirements emerged from three concurrent inputs.

First, document analysis of Volkovich and Avros (2025), which establishes the working corpus (49 Shakespeare-attributed creations from Project Gutenberg), the expected output format (a binary list of suspicious creations), and a published comparison point against which the project’s results can be evaluated.

Second, supervisor consultation through regular project meetings, which shaped many of the methodological decisions documented later in this book: among them the appropriate representation for the per-creation score, the choice between Isolation Forest and simpler thresholding, the handling of short creations, the use of symbolic parameters rather than fixed numerical values, and the addition of multi-run averaging.

Third, the project’s own technical analysis surfaced concerns about possible LLM memorization of Shakespeare, statistical reliability of the delta scores, and the circular contamination prior implied by an Isolation Forest classifier. These concerns motivated the inclusion of a control experiment, the use of multi-run averaging to handle LLM stochasticity, and the switch to Otsu thresholding.

4.3 Research process

The project will proceed in five stages:

- **Stage 1: Proof of Concept.** Implement the pipeline end-to-end on a small subset of the corpus, chosen to include a known mixed-authorship case such as Henry VIII and a confirmed pure-Shakespeare case such as Macbeth. Verify that the delta scores separate authentic from suspicious cases on this subset before scaling up to the full corpus.
- **Stage 2: Reproducibility & sensitivity.** Run M independent runs on identical inputs and verify that per-creation scores are stable across runs. Vary P over a range of values around the pilot default and verify that the classification of every creation is stable across the range.
- **Stage 3: Full corpus run.** Execute the complete pipeline on all 49 creations with a single round (no purification). Produce the sorted scatter plot, apply Otsu thresholding, and record the suspicious set.
- **Stage 4: Optional refinement.** Only if the single-round result proves unstable, re-extract the style profile from the authentic-only set and re-run the pipeline, bounded by $T_{\max}$ iterations. This

purification step is deferred to Phase B and is not part of the core single-round method; if used, the inter-round set differences are reported as evidence of convergence.

- **Stage 5: Control experiment & comparison.** Run the pipeline twice on a sample of creations: with the extracted profile (Mode A) versus with a generic “in Shakespeare’s style” prompt (Mode B). Compare the resulting delta distributions to confirm that the explicit profile contributes beyond the LLM’s memorized prior. Then compare the final suspicious set to that of Volkovich and Avros (2025), reporting both agreement and disagreement cases.

4.4 Expected challenges

- **LLM memorization of Shakespeare.** Modern LLMs have likely been trained on Shakespeare’s creations in their entirety. When asked to rewrite a chunk, the model may regurgitate its memorized version and artificially deflate delta. The control experiment described in Section 4.3 is designed to detect this; if observed, the fallback is a held-out non-canonical corpus.
- **Run-to-run LLM stochasticity.** Even at temperature = 0, LLM APIs are not bit-exact reproducible across calls (non-associative GPU arithmetic, provider-side batching). Mitigation: averaging over M independent runs, explicit model-version pinning, and documented seed handling.
- **Long-context rewrite fidelity.** LLMs rewrite faithfully on shorter inputs but begin to paraphrase as inputs lengthen. The chunk size P is chosen to stay safely in the faithful range. The sensitivity sweep in Stage 2 quantifies this.
- **Profile contamination.** The profile is extracted from the entire 49-creation corpus, including any creations that may not actually be Shakespeare’s, which is exactly what the project is designed to detect. To the extent the corpus contains non-authentic creations, the profile is biased by their inclusion and the threshold τ is calibrated on a contaminated distribution. In the single-round Phase-A method this is a known limitation, mitigated only by the assumption that most of the corpus is authentic; an optional iterative purification (re-extracting from the authentic-only set) is deferred to Phase B and adopted only if the single-round threshold proves unstable.
- **Borderline / non-bimodal distribution.** Mixed-authorship creations (Henry VIII, Pericles) may produce intermediate mean-delta values near τ . If the empirical distribution is not clearly bimodal, Otsu’s threshold is fragile; in that case the histogram is inspected and the fallback is a Gaussian-mixture or manually justified threshold. Borderline creations are reported as a third category even though the classifier itself stays binary.

4.5 Tools and technologies

Table 1 summarizes the principal tools and libraries.

Tool	Purpose
Python 3.11	Primary implementation language.
Pipeline LLM (Gemini 2.5 / GPT-class / Claude)	Style-profile extraction and chunk rewriting; needs a long context window. Final model selected in the POC by measured rewrite fidelity and run-to-run stability.
python-Levenshtein	Function-word edit-distance computation.

Tool	Purpose
scikit-image	Otsu’s threshold algorithm (threshold_otsu); a general 1-D method despite the library’s image-processing focus.
scikit-learn	Optional comparison classifiers (K-medoids, Isolation Forest) for cross-validation against the original paper.
NumPy / Pandas	Numerical and tabular data handling.
Matplotlib / Seaborn	Sorted scatter plots, histograms, threshold visualizations, run-variance diagnostics.
Project Gutenberg	Text corpus (49 creations, identical to the Volkovich and Avros, 2025 corpus).
Git / GitHub	Version control, code/prompt sharing, and submission artifact.

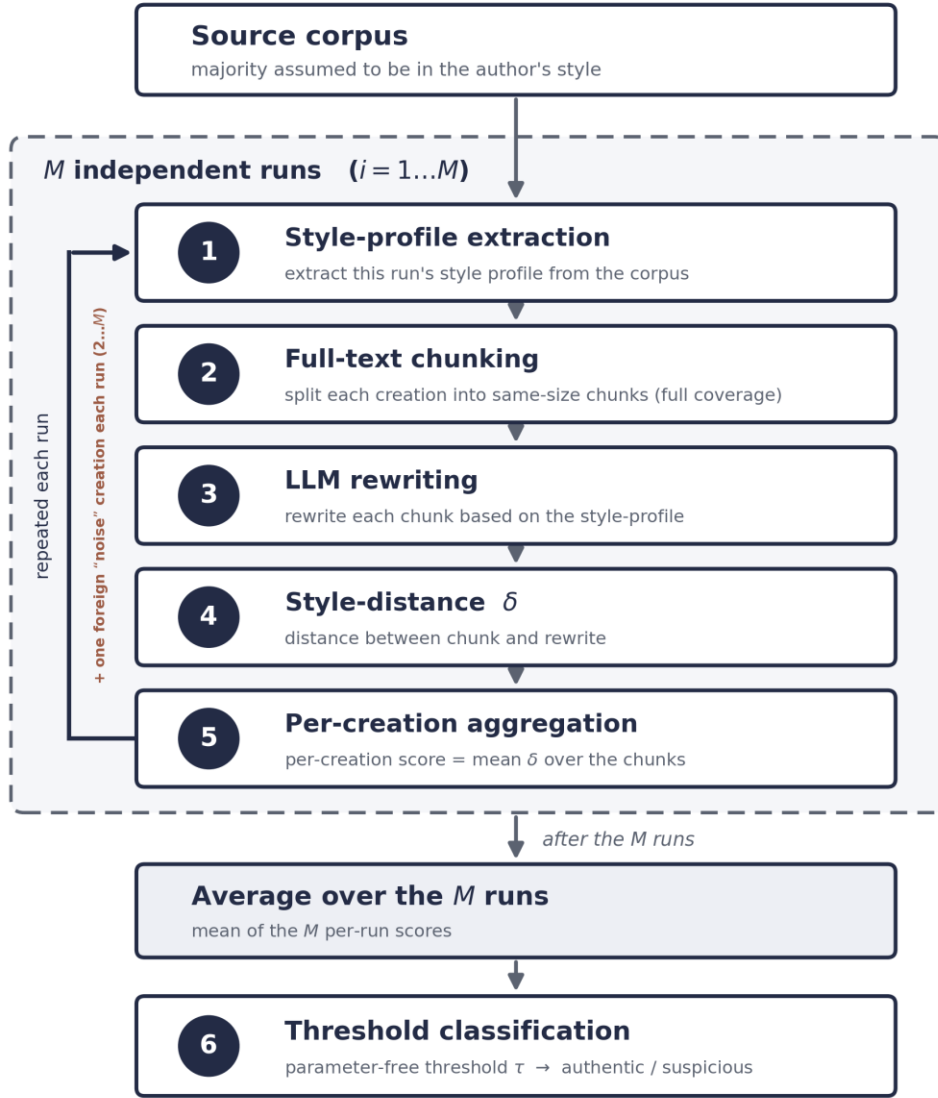
Table 1. Tools and technologies used in the project.

4.6 Solution description

The solution is a six-step pipeline whose per-creation scores are averaged over M independent runs to cancel the LLM’s run-to-run variability. The pipeline takes a corpus of creations as input and produces, for each creation, a binary classification (authentic / suspicious) along with a continuous mean-delta score and a per-creation confidence diagnostic. The procedure is unsupervised, zero-shot with respect to the LLM, and requires no impostor texts or classifier training.

4.6.1 Pipeline overview

The pipeline (Figure 1) consists of six sequential steps, run as M independent runs whose per-creation scores are averaged to cancel LLM stochasticity (Step 5). In the main proposal each run after the first injects a different foreign creation as noise (run 1 uses the plain corpus and noise is non-cumulative), so that a single foreign creation cannot flip the authentic classification; Phase B additionally evaluates a variant with no added noise, since the LLM already varies run-to-run on its own. Full pseudocode appears in Section 4.6.2.



Main proposal: runs 2...M each add one different foreign "noise" creation (run 1: none; non-cumulative).

Phase B also evaluates a no-added-noise variant (the LLM's intrinsic run-to-run variability).

Figure 1. The six-step pipeline, run as M independent runs whose per-creation scores are averaged to cancel the LLM's run-to-run variability.

Step 1, Style-profile extraction. The LLM receives the full text of every creation in the corpus (the majority of the corpus is assumed to be already in the author's style) and is asked to describe the dominant style (vocabulary preferences, pronoun usage, verb forms, sentence structure, punctuation). The prompt does not mention Shakespeare. If the corpus does not fit the LLM's context window in one call, it is split across multiple calls, each holding complete creations (no per-creation truncation).

Step 2, Full-text chunking. Each creation is partitioned into consecutive same-size chunks (default $P = 400$). The last chunk may be shorter. Full coverage, no sampling.

Step 3, LLM rewriting. Within a single run, each chunk is rewritten once under that run’s profile, with instructions to preserve content and change only stylistic features. Across the M runs, each chunk thus receives M rewrites, each paired with an independent profile.

Step 4, Function-Word Edit Distance. For each rewrite, the ordered sequence of function words is extracted from both the original chunk and the rewrite, their Levenshtein distance is computed, and normalized symmetrically by the larger of the two list lengths.

Step 5, Per-creation aggregation. Within each run, the creation-score is the mean of the chunk-deltas. Across the M runs, the final per-creation score is the mean of the M per-run creation-scores. The standard deviation of the M scores is recorded as a per-creation confidence diagnostic.

Step 6, Threshold classification. The 49 averaged per-creation scores are sorted, and a parameter-free 1-D threshold separates authentic from suspicious creations. Otsu’s method is the classifier used in this illustration, one of several (e.g. a Gaussian-mixture split) compared in Phase B. This single-round classification is the Phase-A output.

Single-round method and optional purification. The Phase-A method is single-round: M independent runs, each re-extracting its profile from the full corpus, their per-creation scores averaged, and one Otsu split, with no feedback loop. Because those profiles are calibrated on the full, still-contaminated corpus, an optional iterative purification (re-extracting from the authentic-only set and repeating) could sharpen them, but a wrong early call could also propagate; purification is therefore deferred to Phase B, adopted only if the single-round threshold proves unstable.

4.6.2 Pseudocode

Figure 2 presents the pipeline in pseudocode: M independent runs whose per-creation scores are averaged before a single Otsu classification.

```

Input: corpus  $W$ ; chunk length  $P$ ; number of independent runs  $M$ 
Output: per-creation scores; suspicious set; Otsu threshold  $\tau$ 

for  $i \leftarrow 1$  to  $M$  do                                     ▷  $M$  independent runs
  ▷ main proposal: for  $i \geq 2$ , add one different noise creation (run 1: none; non-
  cumulative)
  profile_i  $\leftarrow$  ExtractStyleProfile( $W$ )                ▷ profile from the whole corpus
  for each creation  $w$  in  $W$  do
    chunks  $\leftarrow$  split  $w$  into consecutive same-size pieces  ▷ full coverage
    for each chunk  $c$  in chunks do
       $r \leftarrow$  Rewrite( $c$ , profile_i)
       $\delta_c \leftarrow$  StyleDistance( $c$ ,  $r$ )                ▷ example: function-word edit distance
      score_i[ $w$ ]  $\leftarrow$  mean( $\delta_c$  over the chunks of  $w$ )
  for each creation  $w$  in  $W$  do
    score[ $w$ ]  $\leftarrow$  mean(score_i[ $w$ ] for  $i = 1 \dots M$ )    ▷ average the  $M$  runs
 $\tau \leftarrow$  OtsuThreshold({ score[ $w$ ] :  $w$  in  $W$  })
suspicious  $\leftarrow$  {  $w$  in  $W$  : score[ $w$ ]  $> \tau$  }
return score, suspicious,  $\tau$ 

```

Figure 2. Pseudocode for the proposed single-round pipeline. P and M are the chunk length and the number of independent runs, respectively.

4.6.3 Function-word edit distance

The distance metric anchors on a fixed list of approximately 120 function words, including both modern English forms (you, has, does, are) and Early Modern English forms (thou, thee, hath, doth, art) so that no a-priori knowledge about Shakespeare’s specific usage is encoded in the metric. For each (chunk, rewrite) pair, the ordered sequence of function words is extracted from each text and their normalized Levenshtein distance is computed:

Formally, given an original chunk c and an LLM-generated rewrite r of that chunk, let $FW(x)$ denote the ordered sequence of function words extracted from a text x . The Function-Word Edit Distance is defined as:

$$\delta(c, r) = \frac{\text{lev}(FW(c), FW(r))}{\max(|FW(c)|, |FW(r)|)}$$

where $\text{lev}(\cdot, \cdot)$ is the Levenshtein edit distance between two sequences and $|\cdot|$ denotes sequence length. The metric lies in $[0, 1]$: it is 0 when the two function-word sequences are identical, and 1 when they share no common function words. The symmetric normalization by the larger of the two lengths makes the metric insensitive to length differences between the original chunk and its rewrite. FWED is the style distance the method is developed with, one of a family of candidates benchmarked in Phase B.

Aggregation to per-creation scores is straightforward. Within a single run $i \in \{1, \dots, M\}$, the score of creation w is the mean per-chunk delta over the chunks of w :

$$s_i(w) = \frac{1}{|\text{chunks}(w)|} \sum_{c \in \text{chunks}(w)} \delta(c, r_i(c))$$

The final per-creation score is the mean of the M per-run scores:

$$\text{score}(w) = \frac{1}{M} \sum_{i=1}^M s_i(w)$$

Distance metric: an example choice among alternatives. The Function-Word Edit Distance is the metric the method is developed with, but because the whole classification depends on the choice of distance, it is treated as one of a family to be compared empirically in Phase B. Alternatives include (i) a function-word frequency-vector distance (cosine or Manhattan over the same word set), order-insensitive and related to Burrows’s Delta, a standard authorship metric; and (ii) a signal-based distance in the spirit of Volkovich and Avros, mapping each text to a numeric signal and comparing the signals to capture rhythm and consistency rather than discrete substitutions. Whether to select the single best-separating metric or to combine several (for example, by averaging their normalized per-creation scores) is itself an empirical question for Phase B.

Because content paraphrasing leaves function words largely intact, the metric captures stylistic change rather than meaning, and it can be read directly as how many of a chunk’s function words the model needed to replace to fit the profile. Authentic Shakespearean chunks should yield low delta values, indicating that the LLM found the chunk already aligned with the corpus’s dominant style; chunks by other authors should yield substantially higher delta values, indicating that more function-word substitution was required to match that style.

4.7 Worked example: Hamlet end-to-end

To make the pipeline concrete, this section walks through its application to a single creation, Hamlet (about 30,000 words). This illustrates the numerical flow from raw text to final classification under the methodology.

Step 1 (profile): the style profile is extracted once per run from the full corpus (a corpus-level step, not a per-creation one) and is shared by every creation scored in that run.

Step 2 (chunking): Hamlet is partitioned into consecutive chunks of $P = 400$ words. With about 30,000 words, this yields $\left\lceil \frac{30,000}{400} \right\rceil = 75$ chunks. The last chunk may be slightly shorter, and no sampling is performed.

Step 3 (LLM rewriting): within a single run, each of the 75 chunks is rewritten once. Across $M = 3$ independent runs (each with its own profile), Hamlet thus receives $75 \times 3 = 225$ LLM rewrite calls.

Step 4 (deltas): for each rewrite, the Function-Word Edit Distance is computed against the corresponding original chunk. For Hamlet, the deltas are typically very low (~ 0.03 to 0.05), since the LLM rarely needs to change function words when the chunk already matches the profile.

Step 5 (aggregation): within each run, Hamlet's score is the mean of its 75 chunk-deltas (about 0.043 in a representative run). After $M = 3$ runs, Hamlet has three per-run scores, for example $[0.042, 0.044, 0.041]$. The final per-creation score is the mean of these three (≈ 0.0423), and the standard deviation across the runs (≈ 0.0015 in this example) is recorded as a confidence diagnostic.

Step 6 (classification): if Otsu's threshold on the 49 averaged scores is about 0.10, then Hamlet, at 0.042, falls well below and is classified as authentic. By contrast, A Yorkshire Tragedy (a play in the Shakespeare Apocrypha) would be expected to produce a mean delta closer to 0.20, placing it above the threshold and classifying it as suspicious.

4.8 Success metrics

Three quantitative success criteria are defined below. These are operational targets set in advance to judge the methodology. They are not measurements already obtained; they are thresholds to be checked against in Phase B, and meeting them would indicate the methodology is performing as intended. The project meets its goal if all three are satisfied at the end of Phase B; specific thresholds may be refined as the methodology develops.

- **Reproducibility.** Classification labels (authentic vs. suspicious) remain stable across the M independent runs for every non-borderline creation; the only creations permitted to vary are the borderline / mixed-authorship cases identified by scholarly consensus (Henry VIII, Pericles, Henry VI Parts I–III, Timon of Athens, as listed in Section 4.1.2). Since that exempt set numbers at most six creations, meeting the criterion implies stable labels for at least 40 of the 49 creations ($\geq 80\%$). Any non-borderline creation whose label varies is flagged and reported with explicit uncertainty annotations. This operationalizes reproducibility at the classification level (what ultimately matters for the project's output) and grounds it in independent literary scholarship rather than an arbitrary threshold on an intermediate numerical quantity.

- **Profile contribution.** The mean delta computed with the explicit profile is statistically significantly lower than the mean delta computed with a generic “in Shakespeare’s style” prompt, evaluated by paired t-test at $\alpha = 0.05$, the conventional statistical significance level. A significant difference confirms the extracted profile contributes information beyond the LLM’s built-in Shakespeare prior.
- **Agreement with consensus.** The pre-registered set of creations for which scholarly consensus on co-authorship is most firmly established (Henry VIII / Fletcher, Henry VI Parts I–III / Marlowe, Timon of Athens / Middleton) is the firmest-consensus subset of the broader borderline group in Section 4.1.2. On this set, the pipeline must correctly classify at least four of these five creations. The “four of five” target allows for one misclassification to accommodate the inherent difficulty of borderline judgments, while still requiring strong agreement on the rest.

4.9 Testing process

Beyond the success-metric checks above, three further tests (together with the noise-robustness check at the heart of the main proposal) will build confidence that the method works as intended.

- **Reproducibility test.** A representative sample of chunks is selected and each is rewritten M times by the LLM with all other parameters held constant. The variance of the resulting deltas is measured. If the coefficient of variation is large on a non-trivial fraction of the test chunks, M is increased and the test is repeated.
- **Synthetic mixture test.** Known Marlowe passages, drawn from outside the 49-creation corpus, are spliced into a small number of pure Shakespeare creations at known positions. The test verifies that (a) the spliced creations are flagged as suspicious by the pipeline and (b) the per-chunk deltas peak at positions matching the spliced-in segments. This is the cleanest ground-truth test available, since the non-authentic positions are known in advance.
- **Held-out generalization test.** A small held-out set of confirmed Shakespeare creations and known non-Shakespeare creations (Marlowe, Jonson, Bacon, drawn from outside the original 49) is reserved before finalizing the pipeline. The finalized pipeline is then run on these held-out creations to verify correct classification on unseen data.
- **Noise-robustness (the main proposal).** The main proposal injects noise: in runs 2... M one different foreign creation (from outside the 49) is added to the corpus before profiling, a fresh intruder each run, while run 1 is left as the plain corpus and noise never accumulates across runs. The per-creation scores are then averaged. Because a single foreign creation should not change the authentic creations’ classification, averaging across runs makes the result robust to contamination. In Phase B the variant with no added noise is also evaluated, relying on the LLM’s intrinsic run-to-run variability (the model already varies on its own); comparing the two shows whether the artificial noise injection is necessary.

All test results, including raw scores, threshold values, classification labels, and diagnostic plots, will be included in the final-phase report and committed to the project’s GitHub repository. Each test will be implemented as an automated pytest suite, so future researchers can re-run the verification with a single command.

4.10 Risk analysis and mitigations

Risk	Likelihood	Impact	Mitigation
LLM memorization inflates accuracy	High	High	Control experiment (Stage 5); fallback to held-out non-canonical corpus.
Run-to-run LLM variance	Medium	Medium	Averaging over M independent runs; pinned model version; logged seeds.
Distribution not bimodal	Low	High	Visual inspection of histogram; fallback to Gaussian-mixture or manual threshold.
Contaminated profile (single round)	Medium	Medium	Profile calibrated on the full, possibly contaminated corpus; stated as a known limitation; optional Phase-B purification if the threshold proves unstable.
Disagreement with paper's results	Medium	Low	Acceptable outcome; report differences as the method's contribution.

Table 2. Risk analysis and mitigation strategies.

5. AI Tools and Prompts Used

Throughout Phase A, the following AI tools were used to support research, methodology refinement, and document preparation. The methodology itself additionally relies on a pipeline LLM as its core component, but that constitutes implementation rather than research support.

5.1 Claude (Anthropic): methodology refinement

Tool homepage: <https://www.anthropic.com/claude>

Claude served as a critical sounding board and as a research and editing assistant: for in-depth discussion and stress-testing of the proposed methodology, related-work search, reflection on how best to incorporate supervisor feedback, and editing and proofreading of the project documentation. All methodological decisions, design choices, and the substantive content of this book are the authors' own. The representative prompts below are lightly polished from their original informal form.

Representative prompt 1 (analysis of the reference paper):

Please analyze the attached paper (Volkovich and Avros, 2025, "Comprehension of the Shakespeare authorship question through deep impostors approach"). For each section, summarize the methodology, identify the assumptions on which it rests, and note any limitations that are relevant to a project that proposes to replace the classifier component with a Large Language Model.

Representative prompt 2 (stress-test of a candidate iterative-purification mechanism):

Below is a draft of an iterative procedure intended to converge on a clean style profile when the initial corpus may include a substantial fraction of misattributed creations. Stress-test it. Identify any unstated assumptions, theoretical weaknesses, and convergence claims that are stronger than what we can actually guarantee. Be skeptical and do not write defensive prose.

Representative prompt 3 (submission requirements coverage):

Cross-check the current draft of the Phase A project book against the Braude 61998 submission instructions. For each required section, indicate whether it is fully covered, partially covered, or missing, and list specifically what would need to be added or revised.

5.2 The pipeline LLM: style extraction and rewriting

Candidate tool homepages: <https://deepmind.google/technologies/gemini/> · <https://openai.com/api/> · <https://www.anthropic.com/claude>

The methodology uses a long-context LLM as the style-extraction and rewriting engine, but does not commit to a specific model in Phase A. It requires only a context window large enough to take whole creations in one profile-extraction call, plus faithful, low-variance rewriting. Candidate models (Gemini 2.5, a GPT-class model, Claude) are compared on the pilot subset during the proof of concept, and the final model is chosen by measured rewrite fidelity and run-to-run stability. The pipeline relies on two prompt templates.

Style-profile extraction prompt (sent once per run, with the full corpus's texts concatenated):

Analyze the dominant writing style in the following creations.

Describe quantitative patterns you observe in:

- Word choice and vocabulary preferences
- Pronoun usage patterns
- Verb forms and conjugations
- Sentence structure and length
- Punctuation habits
- Any other distinctive linguistic features

Be specific and quantitative where possible.

Creation 1: [full text of creation #1]

Creation 2: [full text of creation #2]

...

Chunk rewriting prompt (sent once per chunk per run):

Here is a style profile of a target author:

[the style description from Step 1, current run]

Rewrite the following passage to match this style profile exactly.

Preserve the meaning and content - only transform stylistic features
(word choice, sentence structure, function words, punctuation).

Do not paraphrase or change content words.

Passage to rewrite:

[P words from the creation being tested]

Both prompts are deliberately neutral with respect to Shakespeare: neither contains any reference to Shakespeare, Early Modern English, or any other contextual cue that would bias the LLM toward a memorized prior. This design choice is what makes the methodology unsupervised and generalizable to other authorship questions.

6. References

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